SALES PERFORMANCE ANALYSIS

1.PROJECT OVERVIEW: SALES PERFORMANCE DASHBOARD

This project focuses on designing and developing a comprehensive Sales Performance Dashboard using Microsoft Power BI to analyze, monitor, and evaluate business sales data effectively.

The project integrates **data modeling, DAX calculations, and interactive visualizations** to transform raw transactional data into meaningful business insights. By creating calculated columns, measures, and relationships between tables, the model ensures accurate and dynamic reporting.

2.OBJECTIVE:

The primary objective of this project is to develop an interactive and insightful sales performance dashboard using DAX calculations and advanced data visualizations in Microsoft Power BI.

This project aims to:

1. **Enhance Data Modeling Skills**
    - **Create calculated columns such as *Category Type*, *Revenue*, and *Sales Category*.**
    - **Develop key measures including *Total Sales*, *Order Count*, *Average Profit*, and *Year-to-Date (YTD) Sales*.**
    - **Build relationships between tables to ensure accurate cross-table analysis.**
2. **Analyze Sales and Profit Performance**
    - **Compare actual sales against sales targets by category.**
    - **Evaluate profit margins across sub-categories.**
    - **Identify above-average and below-average sales transactions.**
    - **Measure sales variance and target achievement.**
3. **Identify Trends and Patterns**
    - **Analyze monthly sales trends over time.**
    - **Examine profit vs. quantity relationships using scatter analysis.**
    - **Study geographic sales distribution by city and state.**
4. **Support Data-Driven Decision Making**
    - **Provide clear KPI cards for total sales and targets.**

- **Build a matrix view to compare performance across categories and months.**
- **Present insights through dashboards for better strategic planning.**

5. **Develop Interactive Dashboard Reporting Skills**
    - **Use multiple visual formats (clustered column charts, line charts, donut charts, treemaps, maps, funnel charts, matrix, and cards).**
    - **Create a comprehensive dashboard that communicates sales insights effectively.**

3.PROBLEM STATEMENT:

To design and develop a comprehensive and interactive sales performance dashboard using Microsoft Power BI that:

- Consolidates sales, profit, and target data into a single analytical platform.
- Utilizes DAX calculations to derive meaningful business metrics.
- Enables comparison between actual performance and sales targets.
- Provides time-based and geographic sales insights.
- Enhances decision-making through clear and dynamic visualizations.

4.ATTTRIBUTE (COLUMN/FEATURE)DETAILS:

ATTRIBUTE NAME	DESCRIPTION	DATATYPE
Category	Main product classification (e.g., Electronics, Clothing, Furniture).	Text
Sub_Category	Detailed classification under each main category.	Text
Sales_Category	Sales grouping used for performance comparison or reporting purposes.	Text
Customer Name	Name of the customer who placed the order.	Text
State	State where the order was placed or delivered.	Text
City	City where the order was placed or delivered.	Text

Month Year Sort	Date field used to sort monthly data chronologically in reports.	Date
City State	Combined field of city and state used for geographical analysis.	Text
Country	Country where the transaction occurred.	Text
City Clean	Standardized city name after data cleaning to remove inconsistencies.	Text
Amount	Total sales value for the transaction before profit calculation.	Decimal
Profit	Net profit earned from the transaction after deducting costs.	Decimal
Quantity	Number of units sold in the transaction.	Whole Number
Order ID	Unique identifier assigned to each order.	Whole Number
Order Date	Date on which the order was placed.	Date
Revenue	Total income generated from the sale.	Decimal
Category_Type	Grouping classification of categories	Text
Month Of Order Date	Numeric representation of the order month (1–12) for analysis.	Whole Number
Target	Predefined sales target for a given category or month used for performance comparison.	Decimal

5. TOOLS AND TECHNOLOGIES:

● Excel: Data cleaning, transformation, and Pivot Tables.

● Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. DATA PREPROCESSING:(EXCEL/POWER QUERY)

● Data Cleaning & Transformation done.

CALCULATED COLUMN:

● Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

Category Type = 'Order Details'[Category]&"-"&'Order Details'[Sub-Category]

● Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

REVENUE = 'Order Details'[Amount]*'Order Details'[Quantity]

Create a Calculated Column to Categorize Sales: Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

Sales_category = if('Order Details'[Amount]>AVERAGE('Order Details'[Amount]),"Above Average","Below Average")

Table view

	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales_category
B	₹561.00	212.00	3	Clothing	Saree	Clothing-Saree	1683	Above Average
B-25602	₹119.00	-5.00	8	Clothing	Saree	Clothing-Saree	952	Below Average
B-25603	₹193.00	-166.00	3	Clothing	Saree	Clothing-Saree	579	Below Average
B-25604	₹157.00	5.00	9	Clothing	Saree	Clothing-Saree	1413	Below Average
B-25605	₹75.00	0.00	7	Clothing	Saree	Clothing-Saree	525	Below Average
B-25609	₹25.00	-5.00	4	Clothing	Saree	Clothing-Saree	100	Below Average
B-25610	₹43.00	0.00	3	Clothing	Saree	Clothing-Saree	129	Below Average
B-25611	₹160.00	-59.00	2	Clothing	Saree	Clothing-Saree	320	Below Average
B-25613	₹1,603.00	0.00	9	Clothing	Saree	Clothing-Saree	14427	Above Average
B-25619	₹353.00	90.00	8	Clothing	Saree	Clothing-Saree	2824	Above Average
B-25622	₹534.00	0.00	3	Clothing	Saree	Clothing-Saree	1602	Above Average

CALCULATED MEASURES:

● Calculate Order Count: Define a measure to count the total number of orders in the Order Details table.

Order_Count = SUM('Order Details'[Quantity])

ORDER_COUNT
5615

Calculate Average Profit in Delhi: Create a measure to calculate the average profit for orders placed in Delhi.

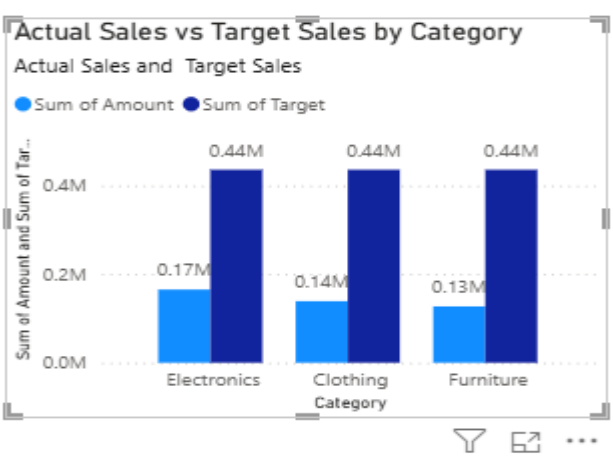
```
Average_Profit_Delhi = calculate(AVERAGE('Order Details'[Profit]),'List of Orders (1)'[City]="Delhi")
```

Calculate Year-to-Date (YTD) Sales: Define a measure to calculate the total sales amount accumulated from the earliest order date up to each order date.

```
YTD_Sales = TOTALYTD([Total_sales],'List of Orders (1)'[Order1 Date])
```

DATA VISUALIZATION:

1. **Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.**



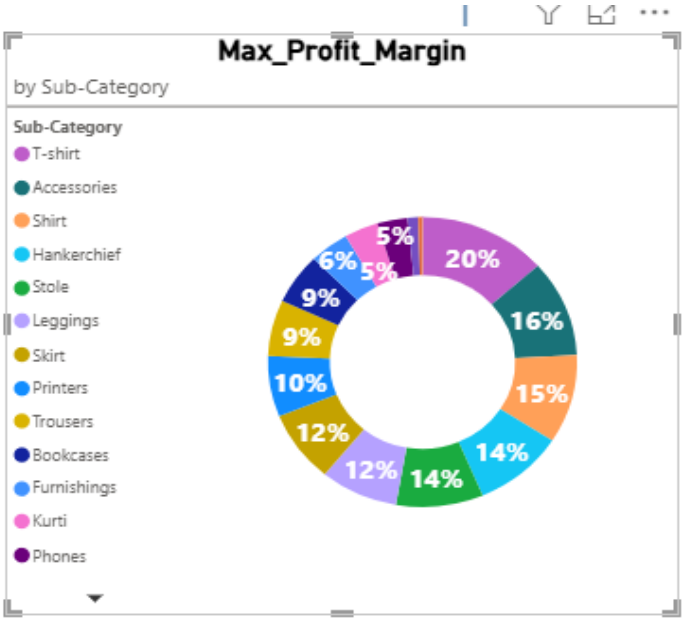
2. **Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each sub-category of products using a donut chart.**

```
Profit_Margin =

DIVIDE(

    SUM('Order Details'[Profit]),
```

```
    SUM('Order Details'[Amount]),
    0
)
Max_Profit_Margin = MAXX(
    VALUES('Order Details'[Sub-Category]),
    [Profit_Margin]
)
```



3.Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.

Step 1: Create a Calculated Column

```
Month Year = FORMAT([Order Date], "MMM YYYY")
```

New column created with only month and Year.

Step 2: Fix Sorting.

Create another column:

```
Month Year Sort = YEAR([Order Date]) * 100 + MONTH([Order Date])
```

Then:

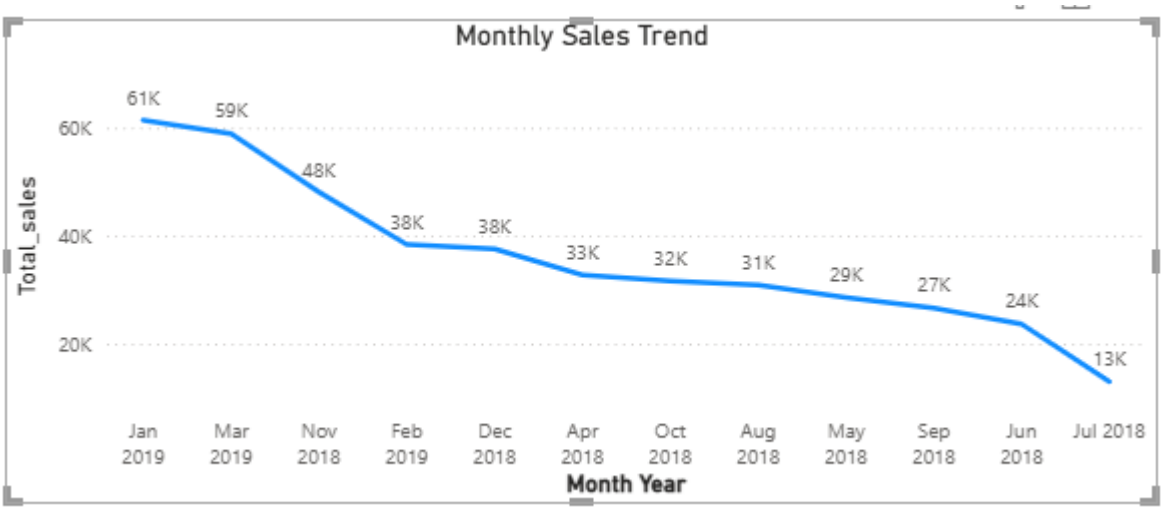
1. Click Month Year column

2. Go to Column Tools
3. Click Sort by Column
4. Select Month Year Sort

Now months will be in correct order.

Then:

- Use Month Year in X-axis
- Use Total Sales in Y- axis
- Set X-axis → Type = Continuous
- Format column as MMM YYYY

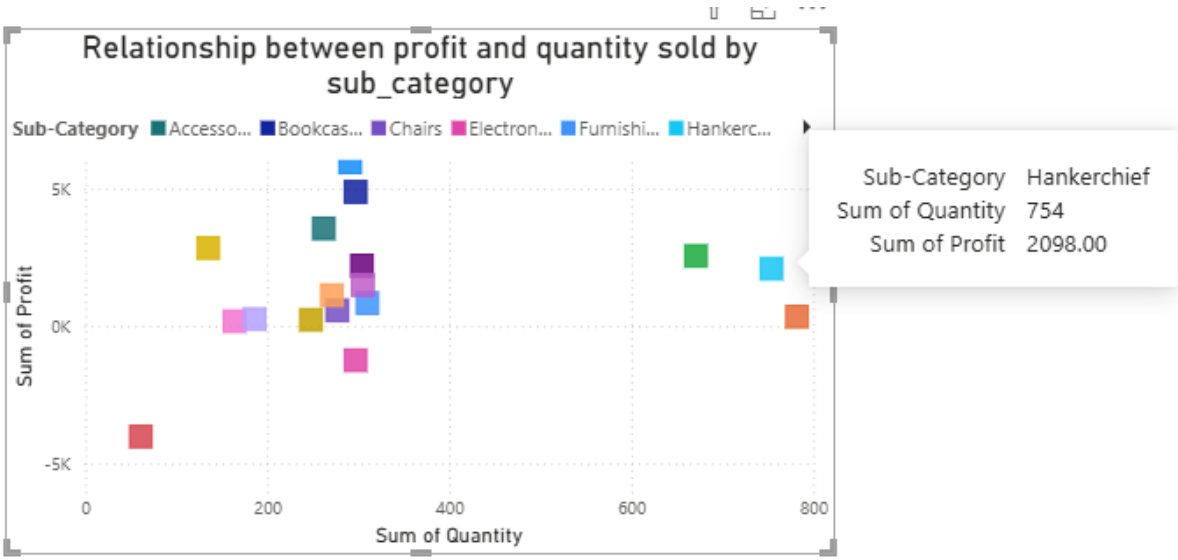


4. Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.

X-Axis→Sum of Quantity,

Y-Axis→Sum of Profit

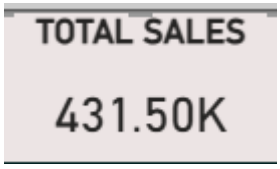
Legend→ Sub Category.



Step 1: Create Required Measures

◆ 1. Total Sales in Order Details Table

Total_sales = sum('Order Details'[Amount])



◆ 2. Total Target in Sales target table.

Total_Sales_Target = sum('Sales target'[Target])

◆ 3. Variance

Sales_Varience = [Total_sales]-[Total_Target]

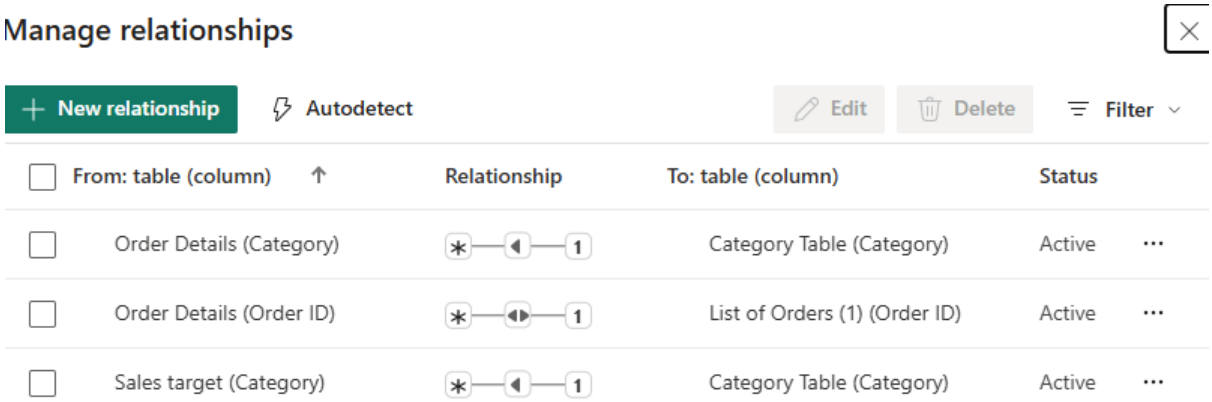
Create new table

category Table = DISTINCT('Order Details'[Category])

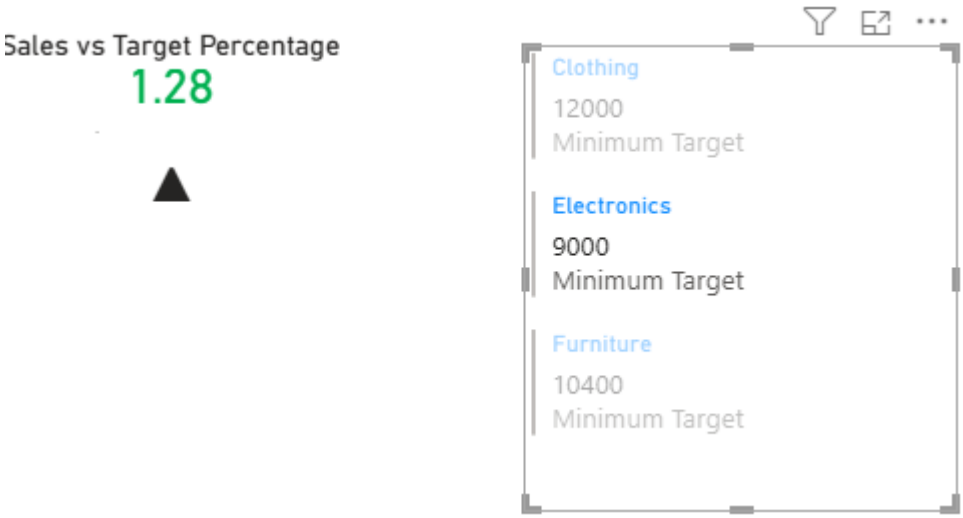
Make a relationship from category table to Order details and Sales Target table .

Form a one to many relationship from category table.

Drag category from newly created Category table to multi row card value field.



5.Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.



6.Sales Performance Matrix:

Build a matrix view to analyze how actual sales compare to

sales targets across different categories and months.

Rows →Category from Category Table

Columns→Month of Order_Date Hierarchy

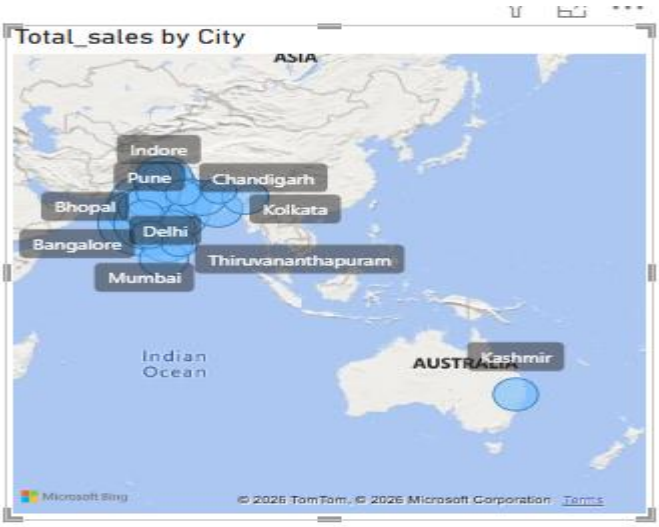
Values →Total Sales,Total Target.

Sales Performance Matrix

Month	January		February		March		April		May		June	
Category	Total_sales	Total_Target	Total_sales	Total_Target	Total_sales	Total_Target	Total_sales	Total_Target	Total_sales	Total_Target	Total_sales	Tot
Clothing	13466	174000	9569	174000	21418	174000	13478	174000	9518	174000	8782	
Electronics	26716	129000	12593	129000	20860	129000	11127	129000	12807	129000	9344	
Furniture	21257	132900	16262	132900	16659	132900	8121	132900	6220	132900	5532	
Total	**61439**	**435900**	**38424**	**435900**	**58937**	**435900**	**32726**	**435900**	**28545**	**435900**	**23658**	

7.Geographic Sales Analysis:

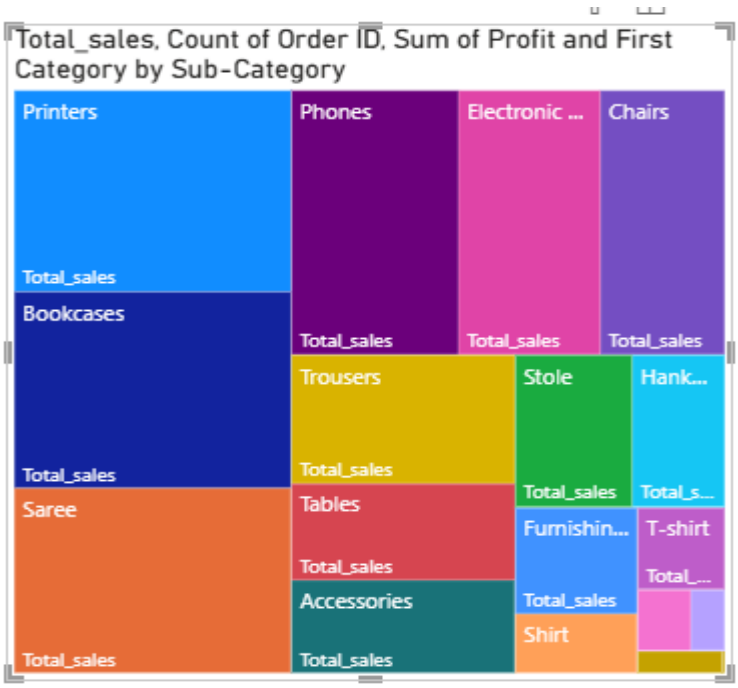
Visualize total sales on a map by city to identify regional sales patterns.



8.Sales Distribution by Sub-Category: Represent the sales distribution across different sub-categories using a treemap.

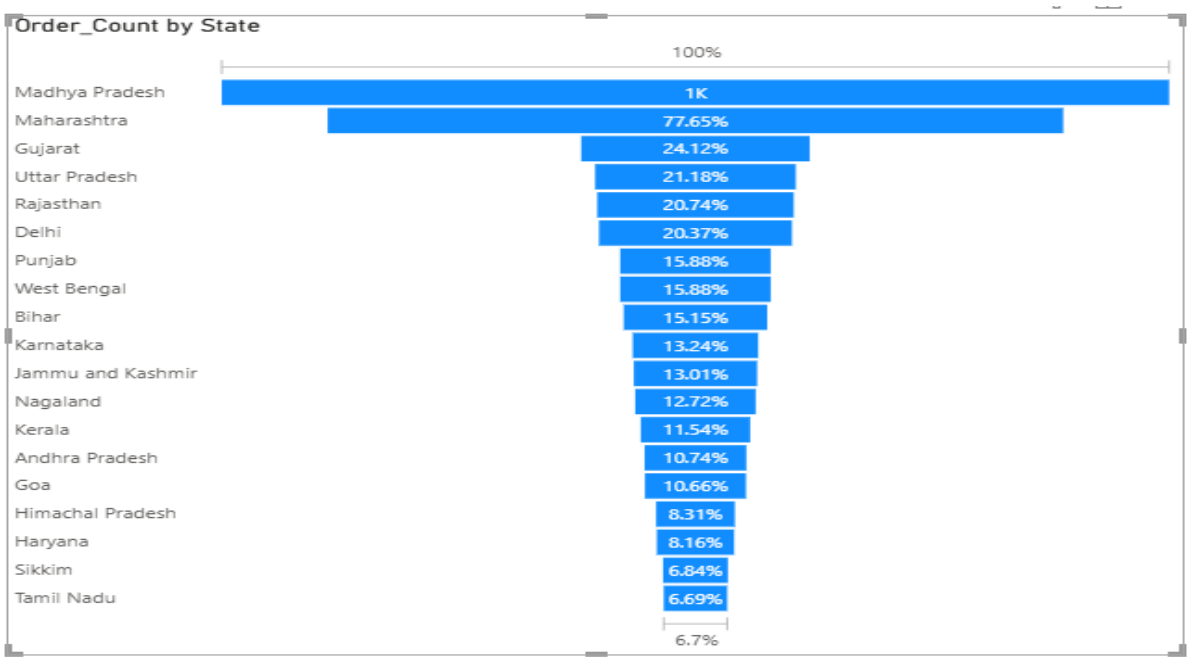
Drag the fields as follows:

- **Category→ Sub-Category, Values → Total Sales, Tooltips → Profit, Quantity, Category**

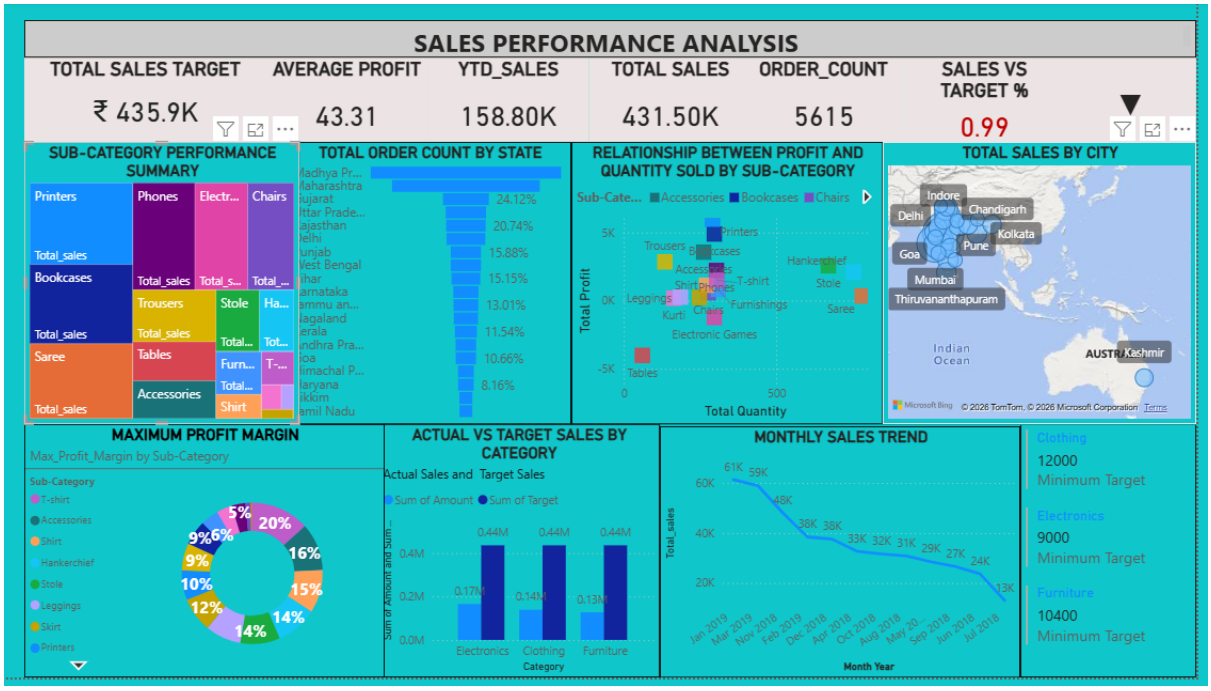


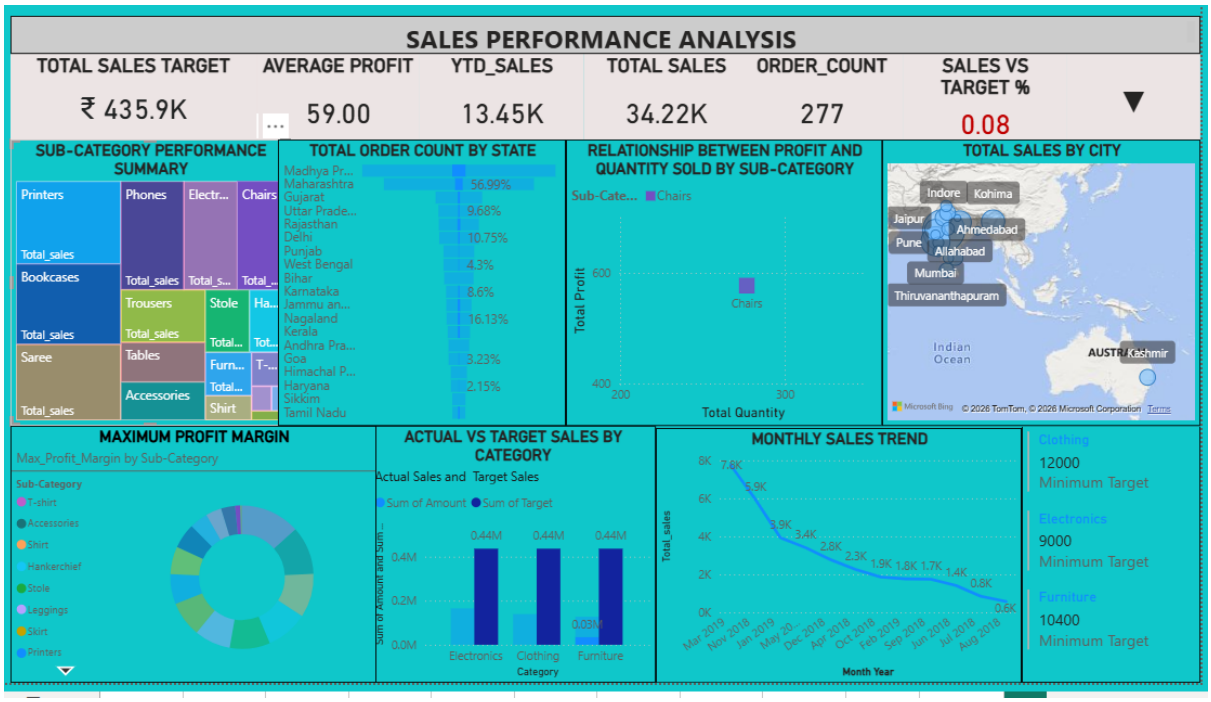
9.Order Count Analysis by State:

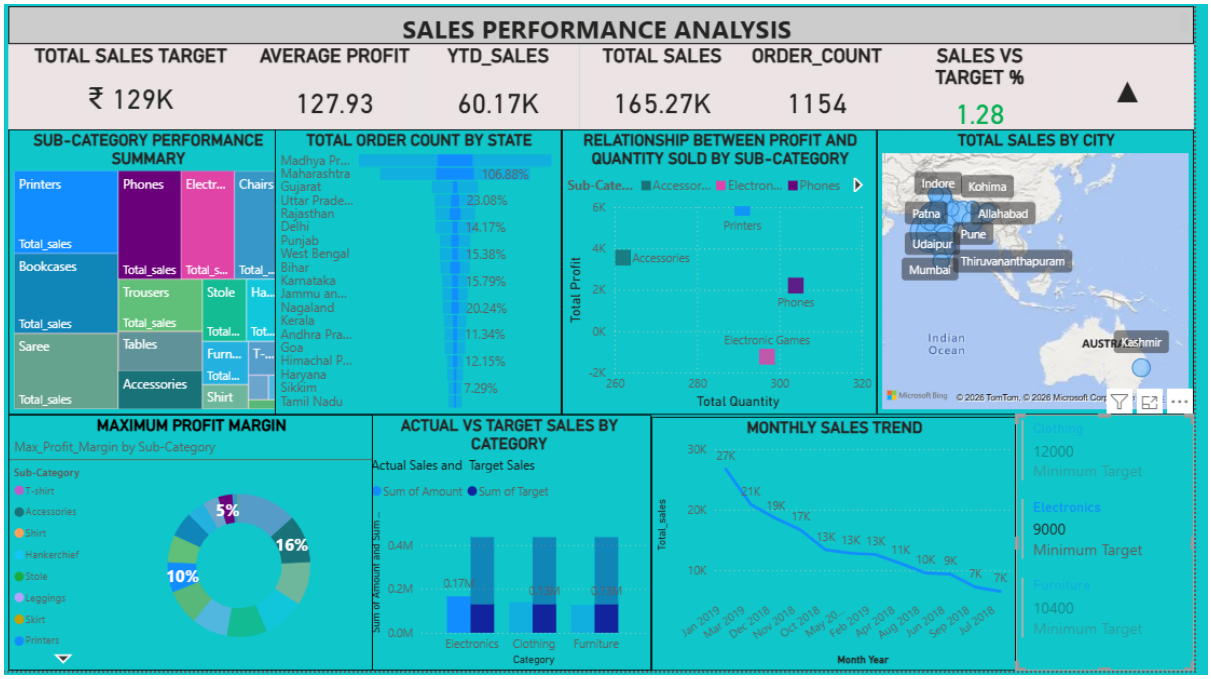
Create a funnel chart to visualize the distribution of order counts across different states.



DASH BOARD


SALES PERFORMANCE ANALYSIS
TOTAL SALES TARGET
₹ 435.9K
AVERAGE PROFIT
43.31
YTD_SALES
158.80K
TOTAL SALES
431.50K
ORDER_COUNT
5615
SALES VS TARGET %
0.99
SUB-CATEGORY PERFORMANCE SUMMARY
TOTAL ORDER COUNT BY STATE
RELATIONSHIP BETWEEN PROFIT AND QUANTITY SOLD BY SUB-CATEGORY
Total Profit
Total Quantity
TOTAL SALES BY CITY
MAXIMUM PROFIT MARGIN
Max_Profit_Margin by Sub-Category
ACTUAL VS TARGET SALES BY CATEGORY
Actual Sales and Target Sales
Sum of Amount
Sum of Target
Electronics
Clothing
Furniture
Category
MONTHLY SALES TREND
Total_sales
Month Year
Clothing
12000
Minimum Target
Electronics
9000
Minimum Target
Furniture
10400
Minimum Target


SALES PERFORMANCE ANALYSIS
TOTAL SALES TARGET
₹ 435.9K
AVERAGE PROFIT
59.00
YTD_SALES
13.45K
TOTAL SALES
34.22K
ORDER_COUNT
277
SALES VS TARGET %
0.08
SUB-CATEGORY PERFORMANCE SUMMARY
TOTAL ORDER COUNT BY STATE
RELATIONSHIP BETWEEN PROFIT AND QUANTITY SOLD BY SUB-CATEGORY
Total Profit
Total Quantity
TOTAL SALES BY CITY
MAXIMUM PROFIT MARGIN
Max_Profit_Margin by Sub-Category
ACTUAL VS TARGET SALES BY CATEGORY
Actual Sales and Target Sales
Sum of Amount
Sum of Target
Electronics
Clothing
Furniture
Category
MONTHLY SALES TREND
Total_sales
Month Year
Clothing
12000
Minimum Target
Electronics
9000
Minimum Target
Furniture
10400
Minimum Target



INSIGHTS:

- **Strong Revenue but Moderate Profitability:**
  The business generated ₹4.31 lakh in total sales, but the overall profit margin remained around 5.55%, indicating scope for improving cost and pricing strategies.
- **Electronics and Clothing Drive Profit:**
  These categories contribute the highest revenue and profit, making them the primary growth drivers of the business.
- **Furniture Has Low Profit Margin:**
  Although Furniture shows reasonable sales volume, its profit contribution is significantly lower, suggesting pricing or cost inefficiencies.
- **Regional Performance Varies Significantly:**
  Madhya Pradesh and Maharashtra are top-performing states in terms of revenue, while some regions show high sales but lower profitability.
- **Seasonal & Monthly Fluctuations Observed:**
  Initial months recorded lower or negative profit trends, followed by gradual improvement, indicating seasonal demand and cost variation impacts.

ANALYSIS

Descriptive Analysis (What Happened?)

- **Total Sales: ₹4,31,502**
- **Total Profit: ₹23,955**
- **Overall Profit Margin: 5.55%**
- **Electronics generated the highest sales revenue**
- **Clothing contributed steady profit**
- **Furniture showed comparatively low profit**
- **Madhya Pradesh and Maharashtra are top-performing states**
- **Initial months recorded lower profits compared to later months**

Diagnostic Analysis (Why Did It Happen?)

- **Furniture category has low profit margin despite good sales**
- **Early months show negative/low profit due to higher discounts or costs**
- **High sales in some regions did not translate into high profit**
- **Seasonal variation affected monthly performance**
- **Profit heavily dependent on Electronics and Clothing categories**

Predictive Analysis (What Is Likely to Happen?)

- **If current growth trend continues, overall profit may increase**
- **Electronics and Clothing likely to remain primary revenue drivers**
- **Furniture may continue generating low profit without pricing changes**
- **Seasonal sales patterns expected to repeat**
- **Meeting increasing monthly sales targets requires steady growth**

Prescriptive Analysis (What Should Be Done?)

- **Improve pricing strategy for Furniture category**
- **Focus marketing on high-profit categories (Electronics & Clothing)**
- **Reduce excessive discounting in low-margin regions**
- **Strengthen performance in underperforming months**
- **Align inventory planning with seasonal demand**
- **Monitor state-wise profitability for better regional strategy**

CONCLUSION:

The Sales Performance Analysis project successfully evaluated the company's sales and profitability using interactive dashboards and data-driven insights. The analysis revealed that while the business generated strong revenue (₹4.31 lakh), the overall profit margin remained moderate at 5.55%, indicating scope for profitability improvement.
Category-wise evaluation showed that Electronics and Clothing are the primary contributors to revenue and profit, whereas Furniture generates comparatively lower profit margins. Regional analysis identified Madhya Pradesh and Maharashtra as key revenue-generating states. Monthly trend analysis highlighted seasonal variations and early-period profit fluctuations.